

Non-linear Optics

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Abstract

In this experiment the nonlinear optical properties of a KDP crystal are studied. They manifest in the creation of photons with doubled wavelength when shining a laser onto the crystal (also called second harmonic generation **SHG**). We start by investigating the SHG intensity with respect to incident laser beam power and relative angle between crystal and laser beam. The power dependence was studied in two manners, **once by rotating the crystal with respect to the incident beam polarization** and also by weakening the laser before it enters the crystal. Both measurements show a **quadratic dependence on power**, with the second method producing more fiducial results. We further observe a **sine-like** dependence on the relative angle between incident laser beam and optical axis of the crystal. Our limited measuring range prevents us from making more founded observations on the behavior. Lastly, we calculate the conversion efficiency by measuring the intensities before and after the crystal. We calculate an efficiency of $\eta = \mathbf{9.0(9) \cdot 10^{-3} \%}$.

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Feedback:

Preparation:

I think you were excellently prepared and new everything I would expect from students taking this course. Very well done!

Grade: 2/2

Experiment:

You were very quick and I had the impression that you carried out the experiment carefully enough to collect good data. However, your report now has me slightly worried that maybe you were a bit too quick. Usually the sinc shape is a bit more symmetric, the power dependency fits really nice, but again the polarization usually shows a large dynamic than what you have measured here.

1.8/2

Report:

Unfortunately there are some bigger issues with your report.

In general your figure caption should be more detailed. Precise language is an issue you should improve on that.

Your introduction and theory part are good. However, a calculation of the phase matching angle would be nice.

In your experiment description you only list things but don't really mention why they are needed or what their purpose is. Science needs to be reproducible so describe exactly what you are doing.

Your results unfortunately remain without any quantitative evidence. Discuss what you have found with numbers and error probabilities.

What do your fit errors mean? E.g. when we expect a quadratic power dependence you need to quantify how likely it is that this is true.

What probability do you find that this is not the case?

Your calculation of the efficiency is wrong. What numbers did you measure for the transmission of the filters and why are they not given?

The values on the PMT are a voltage not an optical power!

Grade: 4/6

Total Grade: 7.8/10

0 Introduction

Lasers are extremely useful tools in many fields of research and offer the realization of great technical applications. They produce coherent light beams of a certain frequency, providing great possibilities in almost all fields of science, including physics, chemistry or medicine. A laser's strength can also be its weakness, that is, they are usually *only* able to emit light of a *fixed* frequency, set by properties intrinsic to the lasing medium. How can this frequency be adapted to be suitable for different applications? We will take a look at second harmonic generation (SHG), an effect that doubles the frequency of a laser beam. In this process two photons are combined to create another photon of twice the energy. However, certain materials and special conditions are prerequisites for SHG to occur.

In this experiment, SHG, also called frequency doubling, is studied in combination with a birefringent **KDP** crystal that converts the wavelength of a HeNe laser of 633 nm to 316.5 nm photons. In the series of measurements, the intensity of the second harmonic generation is measured depending on different quantities. Three series of measurements contain the SHG intensity depending on the incident angle, the laser light polarization, and the laser light power. In the end, the total efficiency of the generation of second harmonic generation is recorded.

This report begins with a short overview over the theoretical concept of second harmonic generation. Then, the experimental setup and the execution of the measurements are described. Afterwards, the results are presented, followed by a brief discussion of the results.

1 Theory

This section describes the relevant theoretical background for the experiment, mainly containing information about SHG [1].

If crystals are exposed to electrical fields, they show a response by the distortion of electrons. For small electric field intensities, this response is proportional to the amplitude of the incident field

$$\vec{P} = \epsilon_0 \chi^{(1)} \vec{E}, \quad (1)$$

where $\vec{P}$ is the induced polarization, ϵ_0 is the vacuum permittivity and $\vec{E}$ is the incident field amplitude. The electrical susceptibility is denoted $\chi^{(1)}$. For higher field amplitudes, the response is not linear. It can be approximated using a Taylor Expansion:

$$\vec{P} = \epsilon_0 \left(\chi^{(1)} \vec{E} + \chi^{(2)} \vec{E}^2 + \chi^{(3)} \vec{E}^3 + \dots \right). \quad (2)$$

The coefficients $\chi^{(i)}$ are tensors, whose elements are determined by the geometry of the crystal and depend on the direction of the incident field. The factors decrease drastically with increasing i , which is why only the first two orders are usually considered. When applying an alternating electric field $\vec{E} = \vec{E}_0 \cos(\omega t)$, a polarization of

$$\vec{P} = \epsilon_0 \left(\chi^{(1)} \vec{E}_0 \cos(\omega t) + \chi^{(2)} \vec{E}_0^2 \cos^2(\omega t) \right) \quad (3)$$

is induced in the crystal. Using $\cos^2(\theta) = 1 - \cos(2\theta)$ this relation can be rewritten as

$$\vec{P} = \epsilon_0 \left(\chi^{(1)} \vec{E}_0 \cos(\omega t) + \frac{1}{2} \chi^{(2)} \vec{E}_0^2 + \frac{1}{2} \chi^{(2)} \vec{E}_0^2 \cos(2\omega t) \right). \quad (4)$$

The polarization is split into three parts: one term representing a constant offset, another represents an oscillation with frequency ω , whilst the third term represents an oscillation with a frequency of 2ω .

Second Harmonic Generation (SHG) is a nonlinear effect, which offers the possibility to double the frequency of photons in a laser beam. A prerequisite is that the second order term $\chi^{(2)}$ is bigger than zero. The basic schema of SHG is shown in Figure 1. While a laser beam with frequency ω enters the crystal, the outgoing beam consists of parts with ω as well as 2ω .

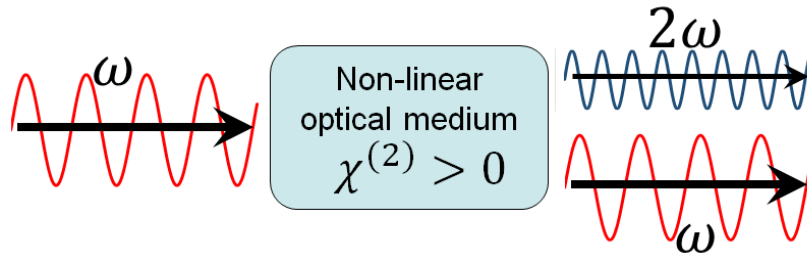


Figure 1: Basic schema of SHG or frequency doubling. Taken from [2].

When the laser beam enters the crystal, it causes a distortion in the delocalized electron cloud. The moving electrons emit electromagnetic radiation as a consequence of their electric

charge. As can be seen in Equation (4), not only photons of the original ω are emitted, but also of the doubled frequency. For light to exit the crystal, constructive interference has to occur throughout the whole crystal. This is only obtained if **the phase matching condition**

$$n_{\omega} = n_{2\omega} \quad (5)$$

is fulfilled. Here n_i refer to the refractive index for the respective frequency. In most crystals this condition cannot be met, as the refractive index grows with increasing frequency. However, birefringent crystals with non-cubic symmetry can fulfill the condition, as their refractive index depends on the polarization. The ordinary refractive index is valid independent of the incident angle relative to the beam axis, while the refractive index of the extraordinary axis depends on the incident angle. This property is used to fulfill phase matching, by aligning the fundamental light with the ordinary axis and the second harmonic generation along the extraordinary axis. By tuning the incident angle θ , phase matching can be achieved, as the refractive index $n(\theta, \lambda)$ is given by

$$\frac{1}{n^2(\theta, \lambda)} = \frac{\cos^2 \theta}{n_o^2(\lambda)} + \frac{\sin^2 \theta}{n_e^2(\lambda)}, \quad (6)$$

with n_o the refractive index of the ordinary axis and n_e of the extraordinary axis.

[the calculation of the phase matching angle would have been nice here.](#)

2 Experimental setup

The main components of the experimental setup are a HeNe laser with a wavelength of 633 nm, a birefringent crystal (KDP), a monochromator and a Photo Multiplier Tube (PMT). Additionally, polarizing optics are used to modify and enhance the setup.

In Figure 2 a picture of the setup with all important components is shown. The laser beam is created by the HeNe laser and afterwards redirected by a mirror. Then it passes a rotatable waveplate, followed by a polarizer, before it is redirected by a mirror again. The beam transverses an objective, then enters the KDP crystal. After leaving the crystal, the beam gets focused by a lens, and enters a monochromator, made with a grating. At the entrance of the monochromator, a filter can be attached, that only lets blue light enter the grating. **The grating is connected** with a voltage source, where around 500 V to 700 V are applied to detect the SHG. The voltage can be adjusted **to optimize the measured voltage at the PMT.** After the monochromator, a Photo Multiplier Tube is used to detect photons.

[I have one major concern with the above paragraph. You describe \(mostly\) correctly what is found in the setup but there is no explanation what these things do! This makes it very hard to judge whether you have understood what you are measuring. Please don't just list things, but explain why they are there and what function they fulfill.](#)

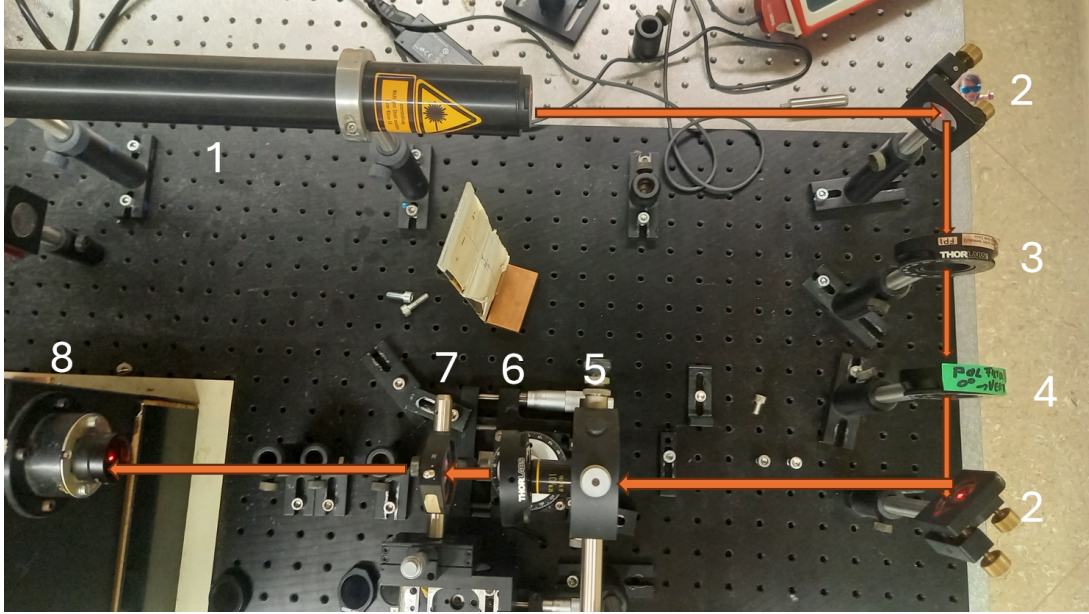


Figure 2: Picture of the experimental setup. Legend: 1) HeNe laser 2) mirrors 3) rotatable waveplate 4) polarizer 5) objective 6) KDP crystal 7) lens 8) monochromator (grating) with attached PMT.

Before the measurements are started, the optical elements have to be adjusted. In the beginning, the second harmonic generation is maximized. To achieve this, the position of the crystal is adapted using the respective screws. In addition, the lens and the waveplate are adapted to maximize the intensity of the beam on the detector.

The first series of measurements deals with the second harmonic generation dependent on the incident angle. The crystal is rotated through a micrometer screw to achieve different incident angles. The translation of the ticks on the micrometer screw to the incident angle is done by using linear interpolation. For every angle used, the intensity is maximized by adjusting the position of the lens located after the crystal.

The second series of measurements aims to find the dependence of the second harmonic generation on the polarization of the incoming laser light. The polarization is varied by rotating the KDP around the ordinary axis with a micrometer screw.

The intensity of the second harmonic generation is then measured as a function of the power of the laser beam. The power is varied by rotating the rotatable waveplate to different angles. For each angle, the second harmonic generation intensity is recorded. Additionally, the power for each angle setting is measured by placing a power meter in the light beam, recording the total power of the laser beam.

The last part of the experiment aims to determine the absolute second harmonic generation efficiency. For this the total output power of the beam has to be detected, including the 633 nm and 316.5 nm contribution. At first, the intensity of the blue light is detected, when the blue light filter is attached before the grating, and the grating is set to wavelengths around 316 nm. To measure the total power, the grating is set to a wavelength of 633 nm. The PMT is very sensitive, so the detector needs to be protected by the use of filters. To reduce the intensity of the red light, three filters are added in the space between the lens and the detector. The

strength of the filters are measured by recording the power of the beam before and after each filter using the power meter.

3 Results

There are multiple dependencies we can study with the experimental setup at hand. We start by investigating the **impact the polarization of the incident beam** has on the converted intensity. The polarization can be changed by rotating the crystal in its mount. We sweep a range of 30° starting around the maximum, which was manually determined beforehand. Using the relation **$P \propto \cos^2(\theta)$** between the power of the incident laser beam and the relative polarization angle **we can plot the recorded intensity of the second harmonic against laser power.** For each angle, three values for the intensity were randomly copied from the oscilloscopes display, which allows us to estimate the uncertainty by computing the standard deviation. The recorded intensities can be seen in Figure 3.

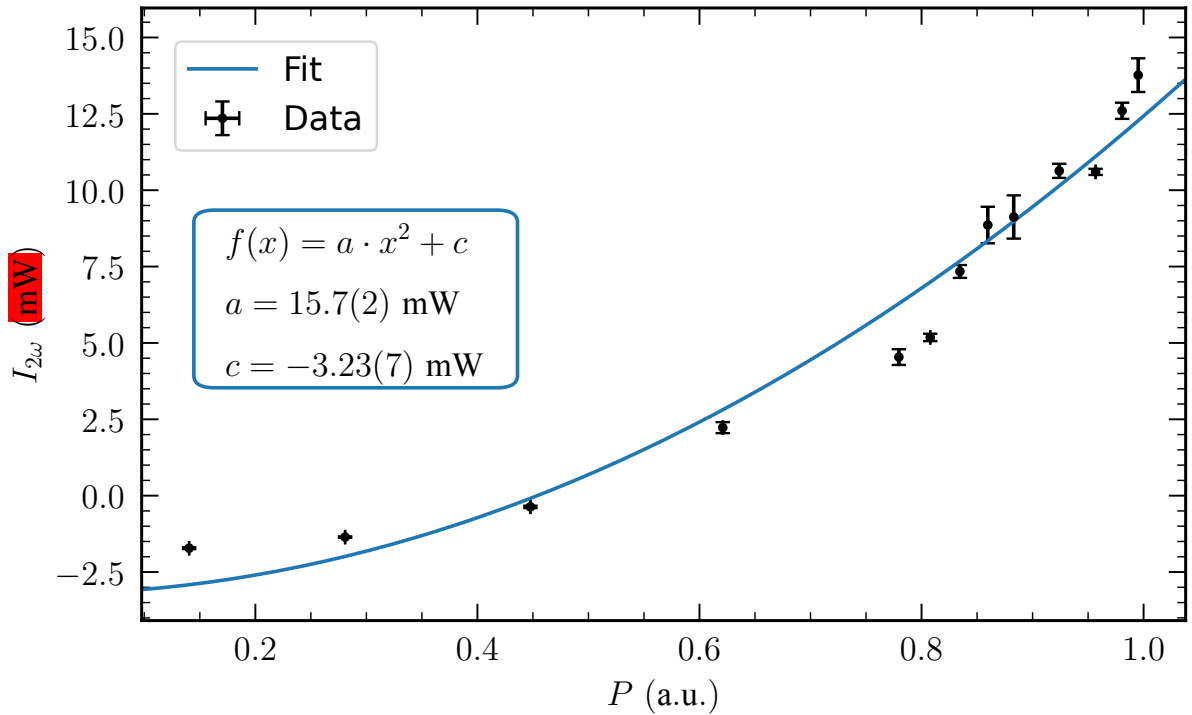


Figure 3: The measured intensity of the second harmonic $I_{2\omega}$ is plotted against incident laser beam power P . The errorbars were estimated by computing the standard deviation of three.

The intensities seem to have a quadratic dependence on the power, which is why we tentatively fitted a parabola of the form $f(x) = a \cdot x^2 + c$ to the data. **The fitted parabola seems to catch the overall trend** but undershoots the data for small powers and overshoots it in the central region of the plot.

We proceed by investigating the dependence on the angle between the crystal (and its optical axes) and the incident beam. For this we start sweeping the full rotating range of the mount starting at one end. For every angle, we optimize the succeeding lens for maximum second harmonic intensity. The recorded intensities are plotted against the angles in Figure 4. Since

we are dealing with diffraction, we also fitted the function $f(x) = a \cdot \text{sinc}(b \cdot x + c) + d$, which can be seen as solid blue line.

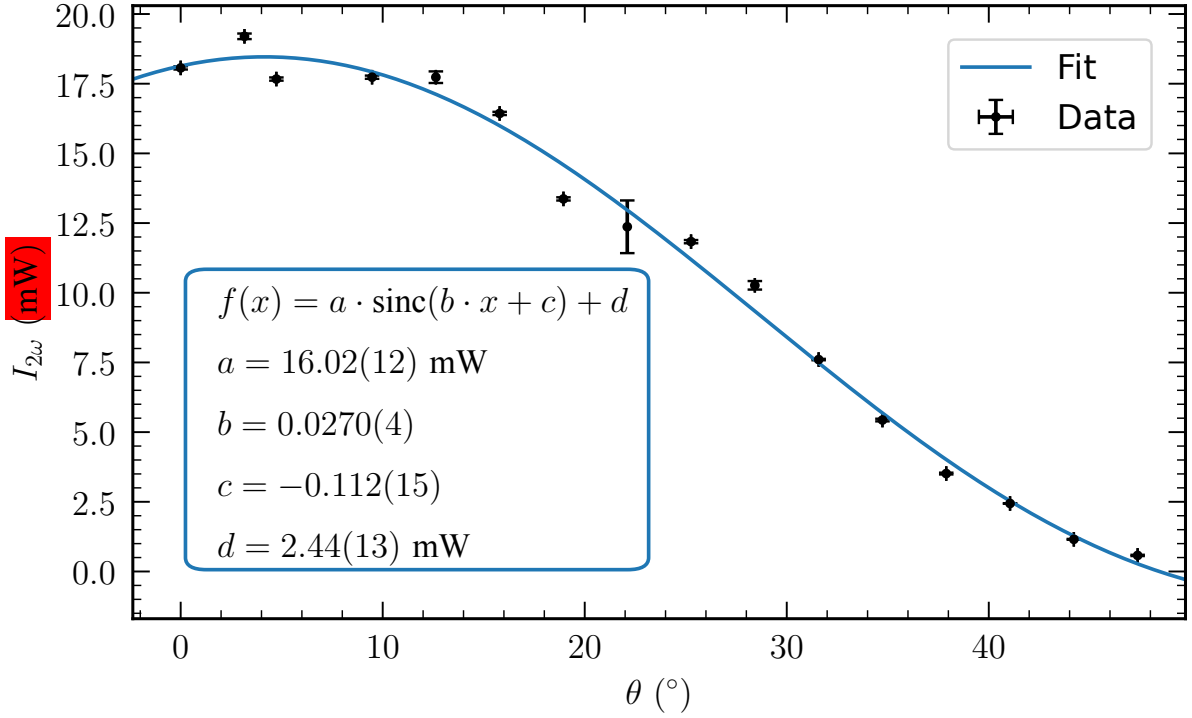


Figure 4: The recorded intensity of the second harmonic is plotted against the angle between crystal and incident laser beam. The whole rotation range was swept, with 0° corresponding to one endstop. A sinc function was fitted to the data and can be seen as solid blue line.

The data seems to visually follow the fitcurve. The characteristic decaying oscillations of the sinc function are outside of our measuring range; to study them, we would need to scan a wider range of angles.

Next, we record the intensity of the second harmonic at different powers of the incident laser beam. We achieve a weakening of laser power by rotating a polarizer through which the beam passes, while measuring with a power meter. We collect datapoints for powers ranging from 0 mW up to a maximum of 7 mW. The data can be seen in Figure 5. Additionally, a parabola of the form $f(x) = a \cdot x^2 + c$, with slope a and offset c was fitted to the data.

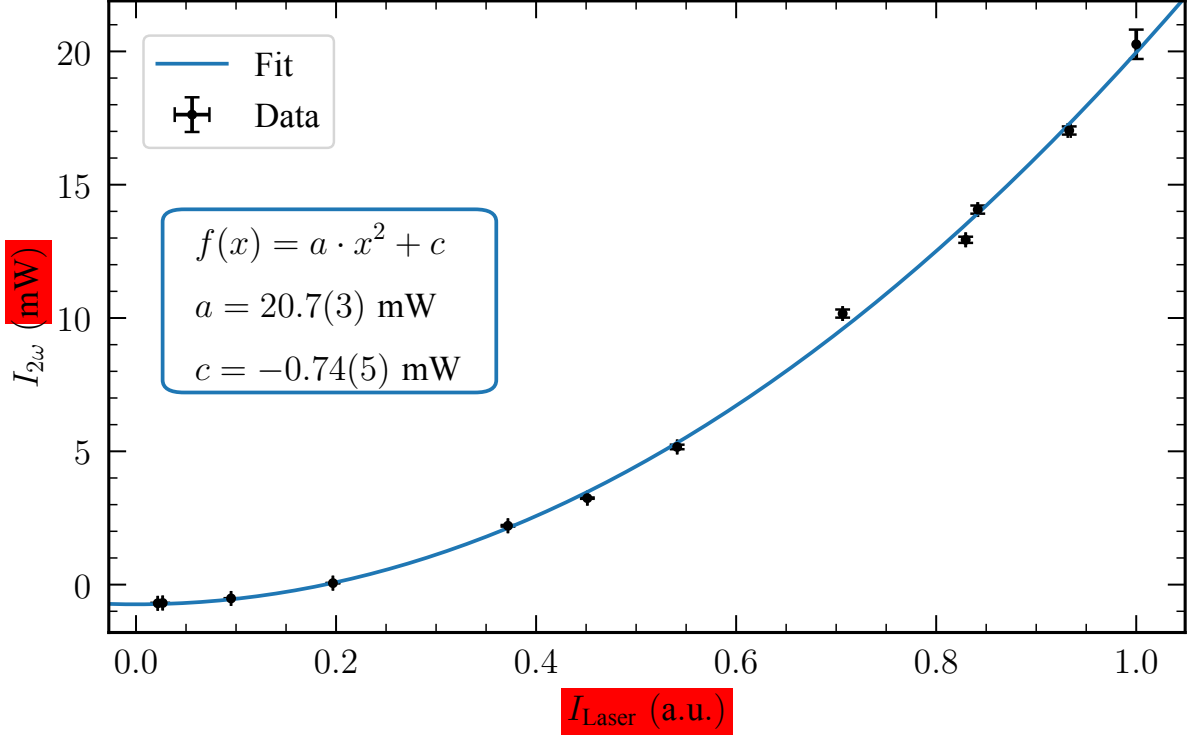


Figure 5: The recoded intensity of the second harmonic is plotted against incident laser power in arbitrary units. A reduction in laser intensity was achieved by rotating the polarizer. A parabola was fitted to the data and can be seen as solid blue line.

The measured intensities in Figure 5 are described well by the parabolic fit.

Lastly, we can calculate the conversion efficiency by recording the intensity of the blue and red laser light. Again, we randomly take note of three values displayed on the oscilloscope to estimate mean and uncertainty for intensity measurements. We record

$$I_{2\omega} = 17.9(3) \text{ mV} \quad \text{and} \quad I_{\omega} = 55.37(6) \text{ mV}$$

for the intensities of the second harmonic ($I_{2\omega}$) and original laser beam (I_{ω}). Before calculating the absolute efficiency η we have to account for the three filters that were placed before the PMT when measuring I_{ω} . For this we characterized all filters by dividing the intensity with and without it. We can then correct I_{ω} by multiplying it with the weakening factor of the filters to get $I_{\omega} = 200(19) \text{ V}$. Now we can calculate the absolute conversion efficiency

$$\eta = \frac{I_{2\omega}}{I_{\omega} + I_{2\omega}} = 9.0(9) \cdot 10^{-5} = 9.0(9) \cdot 10^{-3} \text{ \%}.$$

4 Discussion

In this report we studied the dependence of the second harmonic generation intensity on different parameters. The power dependence was investigated via two different methods; rotating the KDP crystal with respect to the incident beam polarization and weakening the incident beam utilizing rotatable polarizers. Both methods show a quadratic dependence of of SHG intensity on input power, albeit with different qualities. Rotating the crystal has the data roughly following

the expected line with deviations. A likely cause for the observed deviations is the manual rotation of the crystal. This can induce slight misalignments and vibrations, as well as lead to thermal expansion of the metal mount, which all negatively affect the SHG intensity. Weakening the input beam and directly measuring the power before entering the crystal leads data that is in accordance with expectations.

Rotating the crystal along the vertical axis (ie changing the relative angle between input beam and optical axes of the crystal) we observe a sine-like dependence on relative angle within our measuring range, which only covered half an oscillation cycle. We therefore can not comment on exact relation between intensity and angle, only that the intensity falls off with increased relative angle following a sine-like behavior.

The conversion efficiency $\eta = 9.0(9) \times 10^{-3} \%$ is quite small, which means that only a very small percentage of the input photons are converted to photons with doubled frequency. In another experiment, a much higher conversion efficiency of 55.4% could be achieved [3]. However, these results cannot be compared directly, as SHG is a complex effect that depends on many parameters. Under some simplifying assumptions, the conversion efficiency follows the relation

[4]

$$\eta \propto L^2 d_{\text{eff}}^2 P_f / \lambda_f^2 \quad (7)$$

for a nearly collimated beam. In this formula, L is the crystal length, λ_f the fundamental input wavelength, and P_f the fundamental input power. The factor d_{eff} depends on the crystal structure, but is also temperature and wavelength dependent. The relation is not applicable generally, as for high powers, saturation occurs. Nevertheless, it illustrates the dependence of SHG on many parameters. Another import factor that is responsible for a lower conversion efficiency is the hydrophilic nature of the KDP crystal. As the used crystal is already a few years old, it attracts water molecules, which reduce the detected emission of the crystal.

References

- [1] Philipp Schindler, Reza Mosala Nejad. *FPA P07 Non-linear optics*. Universität Innsbruck, Innsbruck, AT, October 2022.
- [2] *Schematic view of the SHG conversion of an exciting wave in a non-linear medium with a non-zero second-order non-linear susceptibility*. Online; Accessed at 07. 11. 2024. URL: https://en.wikipedia.org/wiki/Second-harmonic_generation#/media/File:Schematic_of_the_SHG_conversion_of_an_excited_wave_in_a_non-linear_medium.png.
- [3] V. T. Phan et al. "Fabrication of KDP crystal prisms for second harmonic generation". In: *Optik* 171 (2018), pp. 230–236. ISSN: 0030-4026. URL: <https://www.sciencedirect.com/science/article/pii/S0030402618308362>.
- [4] W. GROSSMAN. *ESTIMATES OF (SHG) SECOND HARMONIC GENERATION EFFICIENCY*. Online; June 14, 2023 . Accessed at November 13, 2024. URL: <https://www.gamdan.com/blog/estimates-of-second-harmonic-generation-shg-efficiency>.

Erklärung

Hiermit versichern wir, dass der vorliegende Bericht selbständig verfasst wurde und alle notwendigen Quellen und Referenzen angegeben sind.



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Alexander Helbok

November 16, 2024

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Date



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Max Koppelstätter

November 16, 2024

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Date